

SENG1050 Assessment 2

Web-Based Assignment 1 (15%)

Due Date: 11:59 pm on 05 Sep (Week 6)

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The objectives of this assignment are as follows:

1. To exercise your ability to create web pages using HTML and CSS.
2. To test your ability to present well-defined information using XML and DTD.

Note that this assignment will be developed further in Assessment 3 (Web-Based Assignment 2).

1. Outline

The introduction of Nintendo Switch 2 has taken the world by storm, with pre-order demands hitting 3.5 million units in the first four days. The board of directors of a 50-year old Newcastle toy store wishes to transform their business by investing in Nintendo Switch games.

With this in mind, the board has invited you, a promising student of the University of Newcastle with web development skills, to develop their website, *FeverGamez*. They would like you to construct a prototype of the web page to be launched. The prototype website should present Nintendo games from TWO genres. Each genre should also have THREE games for purchase. For example, if you choose adventure as a category, you should also think about THREE adventure games for purchase. Note that you can make up your own genre and games (they don't have to be real) - be creative!

Your website should still be simple, friendly and attractive. This can be achieved by satisfying the requirements and completing the tasks in the following sections:

- [Section 2: General Requirements](#)
 - [Section 3: HTML Documents](#)
 - [Section 4: XML Documents](#)
 - [Section 5: CSS](#)
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2. General Requirements

Your website must maintain a proper directory. All files must be stored in a directory named *fevergamez*. For clarity, all images included in the website must also be stored in the *fevergamez/images* directory. You may include additional directories according to your discretion.

3. HTML Documents

For this prototype website, you are required to create three web pages: Homepage, About Us page and Contact page. These pages need to comply to specifications mentioned in the following sections.

3.1 All Web Pages

All the web pages that form your website should satisfy the following:

- Coded according to the HTML Living Standard.
- Contain the full HTML document structure, i.e., include `<!DOCTYPE>`, `<html>`, `<head>`, `<meta>`, `<title>` and `<body>`.
- Contain a navigation bar that allows complete navigation capabilities across your website, such that:
 - Each webpage should contain hyperlinks to every other webpage.
 - The hyperlinks are implemented using *relative* URLs.
- The footer of each page should contain a copyright statement including your name, student number, email address, and degree.

3.2 Homepage

The Homepage should be named `index.html`. This page should contain the following:

- A welcome note section containing:
 - Proper and meaningful level 1 heading.
 - One to two meaningful paragraphs of welcome note.
 - One meaningful blockquote.
- A feature product section containing:
 - Proper and meaningful level 1 and 2 headings.
 - An unordered list or an ordered list of THREE feature products (regardless of genre).
 - One short description paragraph for each product (brief without in-depth details).
 - One image for each featured product.
- One hyperlink to an external website (e.g., <http://www.newcastle.edu.au>) using *an absolute URL* in any of the sections above.
- Correct utilization of EM, STRONG, and ABBR tags in any of the sections above.

3.3 About Us Page

The About Us page should be named `about.html`. This page should contain the following:

- A history section containing:
 - Proper and meaningful level 1 heading.
 - One to two meaningful paragraphs of history.
- A privacy policy section containing:
 - Proper and meaningful level 1 and 2 headings.
 - Paragraphs of discussion on privacy policy (brief without in-depth details). You may use a template such as that at <https://business.vic.gov.au/tools-and-templates/privacy-policy-template/> and refine it using GenAI tools.
- A terms and conditions section containing:
 - Proper and meaningful level 1 and 2 headings.
 - Paragraphs of discussion on terms and conditions (brief without in-depth details). You may use a template such as that at <https://termly.io/resources/templates/terms-and-conditions-template/> and refine it using GenAI tools.

3.4 Contact Page

The contact page should be named `contact.html`. This page allow visitors to send messages to the site owners. This will be done as a HTML form, which sends the information using a POST request to

<https://jkorppela.fi/cgi-bin/echo.cgi> when they click the "Submit" button. The following things need to be included on the form:

- An appropriate level 2 heading
 - The name of the person sending a message
 - An email address for replies
 - The subject of the message
 - The message itself
 - Whether or not the message should be considered urgent (use two radio buttons for this).
 - Include buttons to submit and clear the form
 - All inputs are required, the form should not submit unless the entire form has been filled out.
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4. XML Documents

For this prototype website, you are required to create a XML template and TWO XML data files. XML template is typically required as a placeholder since data that you are hosting on your website may change over time while maintaining a fixed structure. Meanwhile, the XML data files will hold actual data that is relevant when the website is available.

4.1 XML Template With DTD

The XML template should be named `xmltemplate.xml`. The document should be able to store as many games as required within a given genre. You may decide the tags that you wish to use to describe the games, but for each game, you should at least have element tags for the following items:

- Name (only one)
- Description (only one)
- Pricing
- Search tags (zero or more)
- Reviews (zero or more). This tag should have an attribute that indicates the score of the review (from 0 to 5 stars).
- The filename of a picture of the product (one or more, use a relative path)
- URL of a webpage related to the product (one, use an absolute path)

You will also need to write a DTD for your XML template. Generally, it is your choice whether the DTD is internal or external, however, for this assignment, you should make the DTD internal because it is easier to validate your XML and DTD.

Your DTD will need to describe all the tags and attributes that you used in your XML. It will also be important to use optional modifiers (+, *, ?) to indicate how many instances of each tag are allowed (according to what is being specified). Not least, define at least two entities (<!ENTITY>) for use in the XML document.

Finally, note that the template will be a **valid** XML file. Although we are more concerned about the tags (the organization of the data) and **not the actual data** about any games, you may need to include some arbitrary data to allow your page to be validated. For example, if the *default* of an attribute is `#REQUIRED`, the attribute needs a value. Otherwise, it will not be validated.

4.2 XML Data Files

Once you have your XML template with DTD, you will need to create TWO separate XML data files. Each XML data file is created for each of the game genres you identified. They should be named `xmldata1.xml` and `xmldata2.xml` respectively. These XML data files should satisfy the following:

- Each XML data file should contain **data** for THREE games.
 - Each XML data file should contain the internal DTD at the top of the document.
 - The XML data files **MUST** have the same tags and DTD as the XML template.
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5. CSS

In general, in order to draw your target audience, make sure that your website considers factors such as navigability, content layout, element positioning on screen, text size, colour contrast, readability, consistency, and accessibility. All visual formatting of the Web site will be implemented using Cascading Style Sheets (CSS). You will be using external style file, document-level style sheet, and inline styles in this assignment.

Create at least one external style file named `style.css`. This external style file should achieve the following CSS requirements to all your web pages:

- Left and right side margin of at least 1 cm for the main content (body) of all webpages.
- A nice background for the webpages. The text must be readable with the background.
- The text colours of a link before they are visited, after they are visited, and during a user action must be different from each other. The link must appear noticeably bigger when you move the cursor over them.
- Include at least one *id* and one *class* selector.
- Hooked to at least one `<div>`, used to provide some interesting styling (your choice of borders, background, floating alignment, etc.)
- Hooked to at least one `<span>`, used to provide some formatting (e.g., a colour or font change) for a few words of text.

You must also include at least one document-level style sheet and at least one inline style to the Homepage (`index.html`). Note that document-level style sheet is used to apply styles that are unique to the particular page, whereas inline styles should only be used for one-off visual effects.

6. Coding Style

All of your work (HTML, XML, and CSS) should be validated and valid (meaning no errors). Use the following for validation:

- HTML: <https://validator.w3.org>
- XML: <https://www.xmlvalidation.com/>
- CSS: <http://jigsaw.w3.org/css-validator/>

Your HTML, XML, and CSS codes must be neat, clear, and indented. For all code files, you must include comments at the top of the file, stating the name of the file, the name of the author and the date the file was created. You must also at least include comments in your code files which addresses the marking criteria.

You must also include a README file named `README.txt` which includes all of your references.

7. Submission

Submission will be done via [Canvas](#). To submit your files, place the *fevergamez* directory into a `.zip` archive. The `.zip` archive should be named `A2_CXXXXXXXX.zip` where `XXXXXXXX` is replaced by **your** student id. Make sure your *fevergamez* directory contains the following:

- Homepage (`index.html`)
- About Us page (`about.html`)

Contact page (`contact.html`)

- XML template with DTD (`xmltemplate.xml`)
- XML data file1 with DTD (`xmldata1.xml`)
- XML data file2 with DTD (`xmldata2.xml`)
- CSS style file (`style.css`)
- All images used for the website (stored in the *images* directory)
- README file (`README.txt`)

Note that failure to follow submission instructions will incur a penalty. Assignments submitted after the deadline will also be penalized by 10% per day late (including weekend). For example, an assignment worth 20% marked at 78% may be graded as:

- On time: final mark = $0.78 * 1.00 * 20 = 15.6$
- 1 day late: final mark = $0.78 * .9 * 20 = 14.04$
- 3 days late: final mark = $0.78 * .7 * 20 = 10.92$
- 5 days late: final mark = $0.78 * .5 * 20 = 7.8$

Extensions will only be granted with a valid and approved Adverse Circumstances application for the assignment. Adverse Circumstances application should also be submitted at least a day before the due.

Warning - Code Generation Tools

*You are **NOT** allowed to use tools which generate HTML, XML, CSS, PHP, or JavaScript code, either from a visual representation or otherwise. This includes DreamWeaver, Microsoft Frontpage, GenAI tools (e.g., ChatGPT), saving to HTML from MS Word, etc. You must write all code yourself in a **plain-text** editor, such as Notepad++ or VS Code (with Live Server extension only). Any assignments that use code generated from a tool will be given **ZERO**.*

Warning - Plagiarism

Plagiarism is defined as presenting the work of others' as if it was your own. This includes the copying of content from other people's web pages (and other sources) without "appropriate referencing" (indicating the parts of your assignment which have been copied and where they were copied from).

*If you "quote" or use other people's work (or images) in your assignments, then you **must** reference them appropriately (using a README.txt file). Failure to do so will cause you to lose marks for the assignment. An assignment that consists largely of work copied from elsewhere will receive very low (zero) marks, and may be reported to the school for suspicion of plagiarism.*

Warning - GenAI Usage

The use of GenAI tools (such as ChatGPT) is forbidden for the writing of any code. However, generative AI may be used for purely creative aspects of the website (such as coming up with descriptions, welcome text, and images) or wherever explicitly stated. AI generated content should also be referenced appropriately (using a README.txt file). AI detection tools will check each file, and any content with a sufficiently high detection score will be reported to the School for suspicion of plagiarism.

You can find more information on the University's Policy on Student Academic Integrity in the Course Outline and at [Policy Library](#).

8. Marking Guide

The following serves as a rough guide to how many marks are allocated to each section of the assignment. Please refer to the rubrics on Canvas for a complete breakdown.

Total Marks: 100

- [General Requirements](#): 2
- [HTML Documents](#): 47
- [XML Documents](#): 18.5
- [CSS](#): 11.5
- [Coding Style](#): 21
- You may be penalised for excessively poor aesthetic appearance, untidiness, spelling and grammar problems, and not correctly following instructions: [-10]